

DSA 8070 R Session 5: Multivariate Linear Regression

Whitney Huang, Clemson University

Contents

Load the Motor Trend Car Road Tests data	1
Data Manipulation	1
Summarizing the Responses and Predictors	2
Fitting Linear Regression	4
SSCP	6
Estimated Error Covariance Matrix in R	7
Testing if we need “cyl”	7
Confidence/Prediction Intervals	8

Load the Motor Trend Car Road Tests data

```
data(mtcars)
head(mtcars)
```

```
##           mpg  cyl  disp  hp  drat    wt  qsec vs  am  gear  carb
## Mazda RX4      21.0   6  160 110 3.90 2.620 16.46 0   1    4    4
## Mazda RX4 Wag  21.0   6  160 110 3.90 2.875 17.02 0   1    4    4
## Datsun 710      22.8   4  108  93 3.85 2.320 18.61 1   1    4    1
## Hornet 4 Drive  21.4   6  258 110 3.08 3.215 19.44 1   0    3    1
## Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02 0   0    3    2
## Valiant        18.1   6  225 105 2.76 3.460 20.22 1   0    3    1
```

Data Manipulation

```
# Treat number of cylinders as a categorical predictor
mtcars$cyl <- factor(mtcars$cyl)

# Specify the four response variables
vars <- c("mpg", "disp", "hp", "wt")

# Create the multivariate response matrix
Y <- as.matrix(mtcars[, vars])
```

Summarizing the Responses and Predictors

```
# Compute the sample mean of each response variable  
colMeans(Y)
```

```
##      mpg      disp      hp      wt  
## 20.09062 230.72188 146.68750  3.21725
```

```
# Compute the sample standard deviation of each response variable  
apply(Y, 2, sd)
```

```
##      mpg      disp      hp      wt  
## 6.0269481 123.9386938 68.5628685  0.9784574
```

```
# Compute the sample covariance and correlation matrices  
cov(Y)
```

```
##      mpg      disp      hp      wt  
## mpg    36.324103 -633.0972 -320.73206 -5.116685  
## disp -633.097208 15360.7998 6721.15867 107.684204  
## hp    -320.732056 6721.1587 4700.86694  44.192661  
## wt     -5.116685  107.6842  44.19266  0.957379
```

```
cor(Y)
```

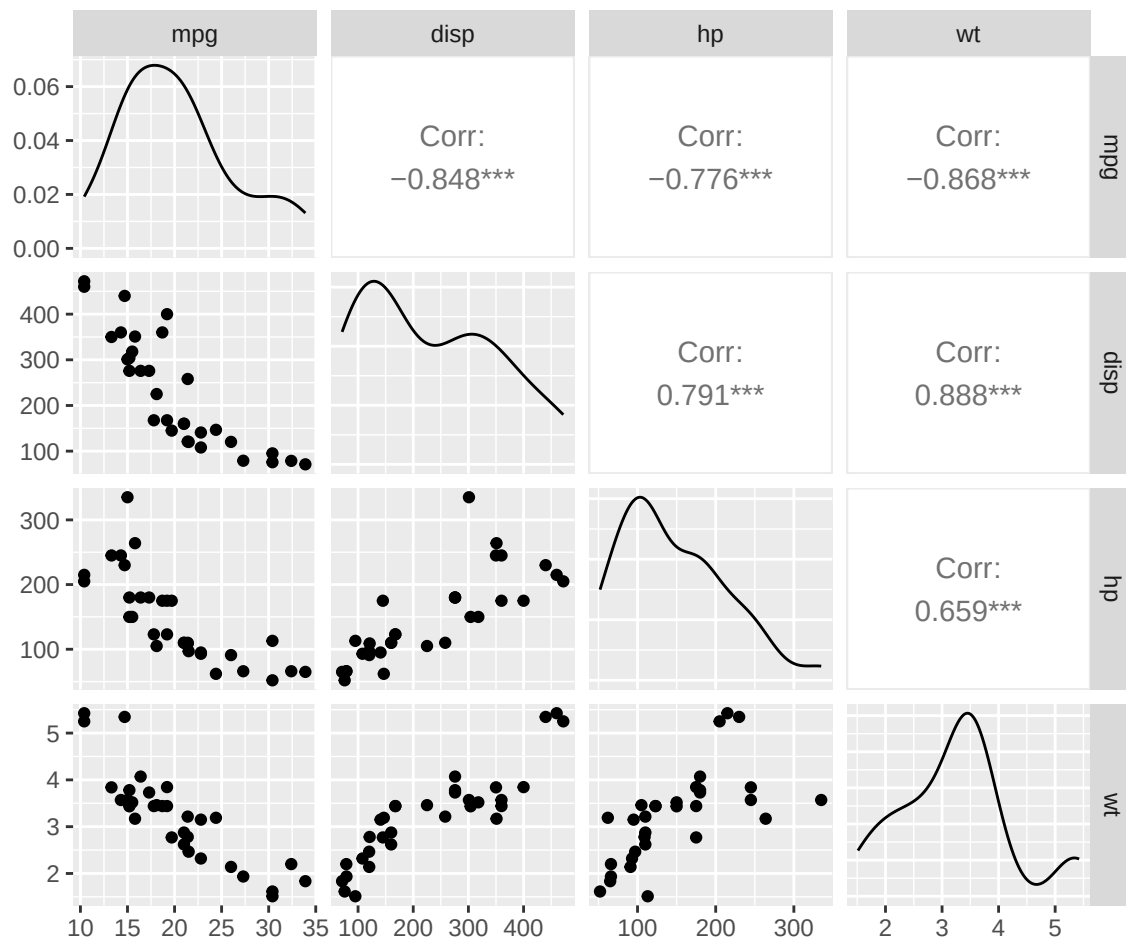
```
##      mpg      disp      hp      wt  
## mpg    1.0000000 -0.8475514 -0.7761684 -0.8676594  
## disp -0.8475514  1.0000000  0.7909486  0.8879799  
## hp    -0.7761684  0.7909486  1.0000000  0.6587479  
## wt    -0.8676594  0.8879799  0.6587479  1.0000000
```

```
# Load the GGally package for creating pairs plots  
library(GGally)
```

```
# Set graphical parameters
```

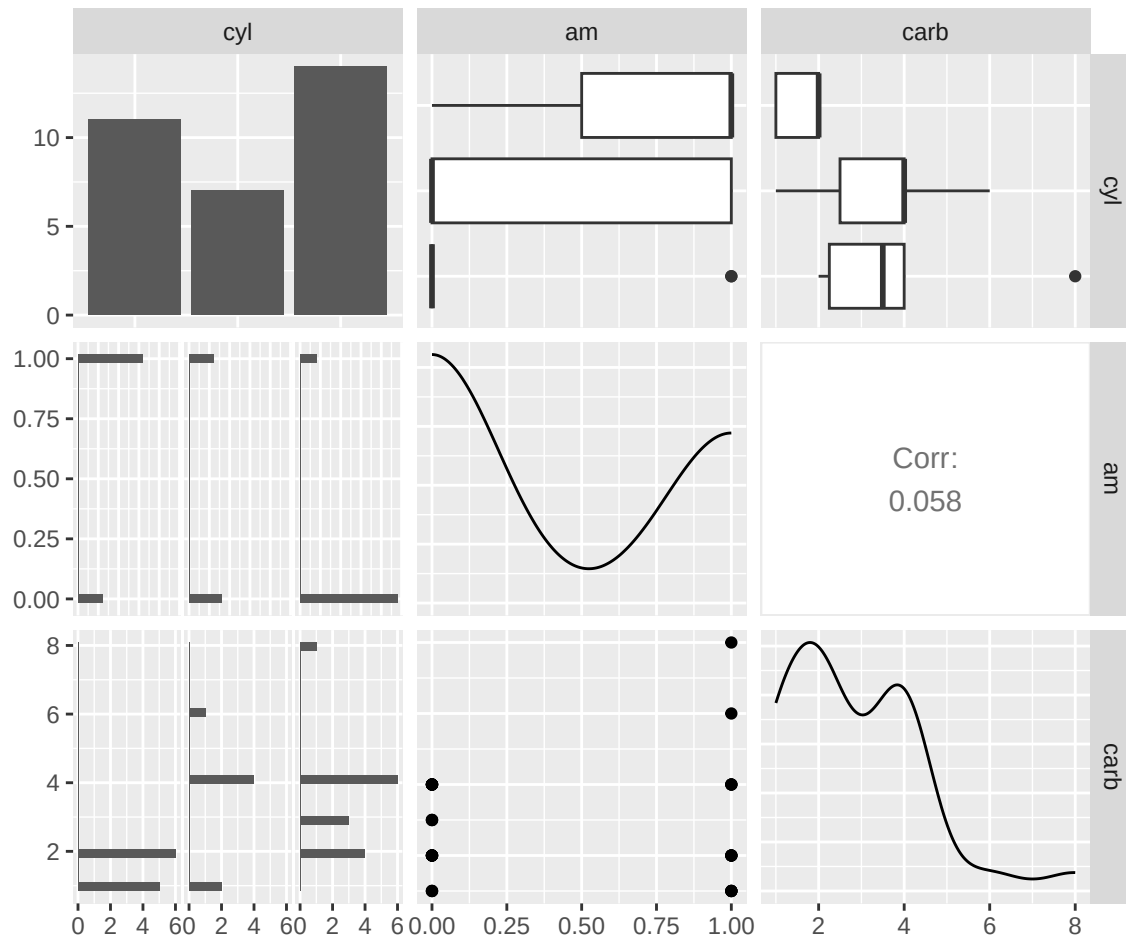
```
par(mar = c(3, 3.5, 0.5, 0.6), mgp = c(2, 1, 0), las = 1,  
    family = "serif")
```

```
# Visualize the marginal distributions and pairwise relationships  
# among the response variables  
ggpairs(as.data.frame(Y), progress = F)
```



```
# Set graphical parameters
par(mar = c(3, 3.5, 0.5, 0.6), mgp = c(2, 1, 0), las = 1,
    family = "serif")

# Visualize the distributions and pairwise relationships
# among the predictor variables
ggpairs(mtcars[, c("cyl", "am", "carb")])
```



Fitting Linear Regression

```
# Fit a multivariate linear regression model with mpg, disp, hp, and wt
# as responses and cyl, am, and carb as predictors
mvlm <- lm(Y ~ cyl + am + carb, data = mtcars)
```

```
# Display coefficient estimates and model summaries for each response
summary(mvlm)
```

```
## Response mpg :
##
## Call:
## lm(formula = mpg ~ cyl + am + carb, data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.9074 -1.1723  0.2538  1.4851  5.4728
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  25.3203     1.2238  20.690 < 2e-16 ***
## cyl6        -3.5494     1.7296  -2.052  0.049959 *
```

```

## cyl8      -6.9046      1.8078     -3.819 0.000712 ***
## am        4.2268      1.3499      3.131 0.004156 **
## carb     -1.1199      0.4354     -2.572 0.015923 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.805 on 27 degrees of freedom
## Multiple R-squared:  0.8113, Adjusted R-squared:  0.7834
## F-statistic: 29.03 on 4 and 27 DF,  p-value: 1.991e-09
##
##
## Response disp :
##
## Call:
## lm(formula = disp ~ cyl + am + carb, data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -82.694 -21.442   0.254  26.500 111.779
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   134.325     21.836   6.152 1.42e-06 ***
## cyl6           61.843     30.860   2.004  0.0552 .
## cyl8          218.991     32.256   6.789 2.72e-07 ***
## am            -43.803     24.086  -1.819  0.0801 .
## carb           1.726       7.768   0.222  0.8258
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 50.05 on 27 degrees of freedom
## Multiple R-squared:  0.858, Adjusted R-squared:  0.8369
## F-statistic: 40.78 on 4 and 27 DF,  p-value: 4.537e-11
##
##
## Response hp :
##
## Call:
## lm(formula = hp ~ cyl + am + carb, data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -41.520 -17.941  -4.378  19.799  41.292
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   46.5201     10.4825   4.438 0.000138 ***
## cyl6           0.9116     14.8146   0.062 0.951386
## cyl8          87.5911     15.4851   5.656 5.25e-06 ***
## am             4.4473     11.5629   0.385 0.703536
## carb          21.2765       3.7291   5.706 4.61e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```
## Residual standard error: 24.03 on 27 degrees of freedom
## Multiple R-squared:  0.893, Adjusted R-squared:  0.8772
## F-statistic: 56.36 on 4 and 27 DF,  p-value: 1.023e-12
##
##
## Response wt :
##
## Call:
## lm(formula = wt ~ cyl + am + carb, data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.66317 -0.34384 -0.03802  0.12334  1.19083
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.76121     0.22133   12.476 1.01e-12 ***
## cyl6          0.19572     0.31280    0.626  0.53675
## cyl8          0.77231     0.32695    2.362  0.02564 *
## am           -1.02547     0.24414   -4.200  0.00026 ***
## carb          0.17491     0.07874    2.222  0.03489 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5073 on 27 degrees of freedom
## Multiple R-squared:  0.7659, Adjusted R-squared:  0.7312
## F-statistic: 22.08 on 4 and 27 DF,  p-value: 3.484e-08
```

SSCP

```
# Compute the sample mean vector of the responses
ybar <- colMeans(Y)

# Get the number of observations (n) and responses (d)
n <- nrow(Y)
d <- ncol(Y)

# Create a matrix with the sample mean vector repeated for each observation
Ybar <- matrix(ybar, n, d, byrow = TRUE)

# Compute the total sum of squares and cross-products (SSCP) matrix
SSCP.Tot <- crossprod(Y - Ybar)

# Compute the regression (explained) SSCP matrix
SSCP.Reg <- crossprod(mvlm$fitted.values - Ybar)

# Compute the error (residual) SSCP matrix
SSCP.Err <- crossprod(Y - mvlm$fitted.values)

# Display the total SSCP matrix
SSCP.Tot
```

```
##          mpg      disp          hp      wt
## mpg      1126.0472 -19626.01 -9942.694 -158.61723
## disp -19626.0134 476184.79 208355.919 3338.21032
## hp      -9942.6938 208355.92 145726.875 1369.97250
## wt       -158.6172  3338.21   1369.972   29.67875
```

```
# Verify the SSCP decomposition: SSCP.Tot = SSCP.Reg + SSCP.Err
SSCP.Reg + SSCP.Err
```

```
##          mpg      disp          hp      wt
## mpg      1126.0472 -19626.01 -9942.694 -158.61723
## disp -19626.0134 476184.79 208355.919 3338.21033
## hp      -9942.6937 208355.92 145726.875 1369.97250
## wt       -158.6172  3338.21   1369.972   29.67875
```

Estimated Error Covariance Matrix in R

```
# Determine the number of predictors, excluding the intercept
p <- nrow(coef(mvlm)) - 1

# Estimate the residual covariance matrix using the error SSCP matrix
SigmaHat <- SSCP.Err / (n - p - 1)

# Display the estimated residual covariance matrix
SigmaHat
```

```
##          mpg      disp          hp      wt
## mpg      7.8680094 -53.27166 -19.7015979 -0.6575443
## disp -53.2716607 2504.87095 425.1328988 18.1065416
## hp      -19.7015979 425.13290 577.2703337 0.4662491
## wt       -0.6575443  18.10654  0.4662491  0.2573503
```

```
# Convert the residual covariance matrix to a correlation matrix
cov2cor(SigmaHat)
```

```
##          mpg      disp          hp      wt
## mpg      1.0000000 -0.3794645 -0.29233405 -0.46209388
## disp -0.3794645  1.0000000  0.35354314  0.71314929
## hp      -0.2923340  0.3535431  1.00000000  0.03825304
## wt       -0.4620939  0.7131493  0.03825304  1.00000000
```

Testing if we need “cyl”

```
# Fit the reduced multivariate regression model without the cyl predictor
mvlm0 <- lm(Y ~ am + carb, data = mtcars)

# Test the joint contribution of cyl across all responses using Wilks' Lambda
anova(mvlm, mvlm0, test = "Wilks")
```

```
## Analysis of Variance Table
##
## Model 1: Y ~ cyl + am + carb
## Model 2: Y ~ am + carb
##   Res.Df Df Gen.var.    Wilks approx F num Df den Df    Pr(>F)
## 1      27      29.862
## 2      29  2  43.692 0.16395    8.8181      8    48 2.525e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Test the joint contribution of cyl across all responses using Pillai's Trace
anova(mvmlm, mvmlm0, test = "Pillai")
```

```
## Analysis of Variance Table
##
## Model 1: Y ~ cyl + am + carb
## Model 2: Y ~ am + carb
##   Res.Df Df Gen.var. Pillai approx F num Df den Df    Pr(>F)
## 1      27      29.862
## 2      29  2  43.692 1.0323    6.6672      8    50 6.593e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Confidence/Prediction Intervals

```
# Specify a new observation with 6 cylinders, manual transmission,
# and 4 carburetors
newdata <- data.frame(cyl = factor(6, levels = c(4, 6, 8)),
                      am = 1, carb = 4)

# Obtain confidence intervals for the mean responses at the new predictor values
predict(mvmlm, newdata, interval = "confidence")
```

```
##      mpg      disp      hp      wt
## 1 21.51824 159.2707 136.985 2.631108
```

```
# Obtain prediction intervals for individual responses at the new predictor values
predict(mvmlm, newdata, interval = "prediction")
```

```
##      mpg      disp      hp      wt
## 1 21.51824 159.2707 136.985 2.631108
```

R does not yet have the capability to produce CIs/PIs with multivariate responses!!

Below is R function to calculate the Multivariate Regression CIs and PIs (taken from Prof. Helwig at U of Minnesota)

```
# Define a function for simultaneous multivariate confidence or prediction intervals
pred.mlm <- function(object, newdata, level = 0.95,
                     interval = c("confidence", "prediction")){
```



```

# Extract the regression formula and construct the design matrix
# for the new predictor values
form <- as.formula(paste("~", as.character(formula(object))[3]))
xnew <- model.matrix(form, newdata)

# Compute fitted response values at the new predictor values
fit <- predict(object, newdata)

# Extract the response matrix and design matrix from the fitted model
Y <- model.frame(object)[, 1]
X <- model.matrix(object)

# Determine the sample size, number of responses, and number of predictors
n <- nrow(Y)
d <- ncol(Y)
p <- ncol(X) - 1

# Estimate the residual variance for each response
sigmas <- colSums((Y - object$fitted.values)^2) / (n - p - 1)

# Compute the variance contribution from estimating the mean response
fit.var <- diag(xnew %*% tcrossprod(solve(crossprod(X)), xnew))

# Add the variance of a new observation for prediction intervals
if(interval[1] == "prediction") fit.var <- fit.var + 1

# Compute the simultaneous multivariate critical-value adjustment
const <- qf(level, df1 = d, df2 = n - p - d) *
  d * (n - p - 1) / (n - p - d)

# Compute the variance for each response at each new predictor value
vmat <- (n / (n - p - 1)) * outer(fit.var, sigmas)

# Construct the lower and upper simultaneous interval bounds
lwr <- fit - sqrt(const) * sqrt(vmat)
upr <- fit + sqrt(const) * sqrt(vmat)

# Organize output for a single new observation
if(nrow(xnew) == 1L){
  ci <- rbind(fit, lwr, upr)
  rownames(ci) <- c("fit", "lwr", "upr")
}

# Organize output for multiple new observations
} else {
  ci <- array(0, dim = c(nrow(xnew), d, 3))
  dimnames(ci) <- list(1:nrow(xnew), colnames(Y),
    c("fit", "lwr", "upr"))

  ci[, , 1] <- fit
  ci[, , 2] <- lwr
  ci[, , 3] <- upr
}

# Return fitted values and interval bounds

```

```

    ci
  }

# Compute simultaneous confidence intervals for the mean response vector
pred.mlm(mvml, newdata)

```

```

##           mpg      disp          hp      wt
## fit 21.51824 159.2707 136.98500 2.631108
## lwr 16.65593  72.5141  95.33649 1.751736
## upr 26.38055 246.0273 178.63351 3.510479

```

```

# Compute simultaneous prediction intervals for a new response vector
pred.mlm(mvml, newdata, interval = "prediction")

```

```

##           mpg      disp          hp      wt
## fit 21.518240 159.27070 136.98500 2.6311076
## lwr  9.680053 -51.95435  35.58397 0.4901152
## upr 33.356426 370.49576 238.38603 4.7720999

```